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SDC
Hackathon
Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	16.0
Project / Solution Name	ShiftHappens
Name (Leader)	**WONG PYNVEN**
Email & Phone (Leader)	**pynven323@gmail.com & 016-3266323**
University Affiliation(s)	**University of Malaya**
Live Prototype URL	https://sdc-hackathon-2026-team-160.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/jenwennnn/SDC_Hackathon_2026_Team16.0

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

ShiftHappens is a single Python web application built with Streamlit and deployed on Replit.

1. User interface: Streamlit pages plus custom CSS and JavaScript (intro animation, step dock, command palette) run in the user's browser.
2. Input: the user uploads or pastes a CV (PDF, DOCX or TXT, read with PyPDF2 and python-docx), picks one of five job templates, or re-uploads a saved profile file.
3. Privacy filter: on the server, regular expressions remove emails, phone numbers and Malaysian IC numbers before any external call.
4. AI step: only the cleaned CV text is sent to the Google Gemini API (google-generativeai SDK; model set by LLM_MODEL, currently gemini-3.8-flash), which returns skills, current role and daily tasks as JSON. Skills are fuzzy-matched (thefuzz, score 80 or higher) to a local ESCO skill list.
5. Logic layer: deterministic Python modules - trend scoring (signals_logic, trend_bridge), role matching (matcher), learning-path planning (pathway) and a fairness test (fairness) - read local JSON data files. No database is used and no profile is stored on the server; users download their own profile as JSON.
6. Hosting: Replit Autoscale deployment; the API key is kept in Replit Secrets.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	**Purpose in Project**	**License**
Python 3.12 (Replit) / 3.13 (local)	Programming language for the whole application	PSF License
Streamlit 1.66.1	Web UI framework and app server	Apache License 2.0
pandas (unpinned, latest at install)	Tables and editable skill table on the Review step	BSD 3-Clause
google-generativeai (unpinned)	Python client for the Google Gemini API (CV extraction)	Apache License 2.0

Name & Version	Purpose in Project	License
thefuzz (unpinned)	Fuzzy matching of extracted skill names to ESCO skill names	MIT
python-docx (unpinned)	Reading text from uploaded .docx CVs	MIT
PyPDF2 (unpinned)	Reading text from uploaded PDF CVs	BSD 3-Clause
python-dotenv (unpinned)	Loading environment variables from a local .env file during development	BSD 3-Clause
Geist font (Vercel)	User interface typography	SIL Open Font License 1.1
Material Symbols (bundled with Streamlit)	Icons in the step dock and buttons	Apache License 2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (Google)	Extracts skills (level, evidence, confidence), current role and daily tasks from CV text after emails, phone numbers and IC numbers are removed. Called only when a user submits a CV. This is the only external API called by the app.	Free tier (team's own API key)
None	-	-
None	-	-
None	-	-
None	-	-

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts the live prototype. Autoscale deployment (deploymentTarget = cloudrun in .replit), run command: streamlit run app.py.	Sponsor-provided Replit credits (USD 25) / Region managed by Replit
Replit Secrets (Replit)	Stores LLM_API_KEY and LLM_MODEL as environment variables, so no key is in the code.	Included with Replit (paid from the same credits)

Service & Provider	**Purpose in Project**	**Plan / Region**
GitHub (GitHub, Inc.)	Source code hosting and version control (Git).	Free
None	-	-
None	-	-

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Google Gemini, model gemini-3.8-flash (set by LLM_MODEL in Replit Secrets; code default gemini-2.5-flash)	Google, via Gemini API	Pre-trained, used as-is through the API (no fine-tuning or training)	Gemini API Additional Terms of Service and Google APIs Terms of Service
No other model. Matching, trend scoring and learning plans use fixed formulas, not AI models.	-	-	-
None	-	-	-
None	-	-	-
None	-	-	-

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	**Source / URL**	**Type (public / private / self-collected / synthetic)**	**License**	**Personal Data? (Y/N)**
ESCO skills, occupations and occupation-skill relations (v1.2.1, downloaded 2026-10-23)	European Commission, https://esco.ec.europa.eu	Public	Free reuse with attribution under the European Commission reuse policy	N
ShiftHappens ESCO subsets: esco_skills_seed_v2.json, future_roles.json (10 roles), occupation_templates.json (5 templates)	Built by the team from the ESCO files above	Public (derived)	ESCO terms (attribution to ESCO / European Commission)	N
Trend signals: signals.json (Synthetic demo dataset v1, 3 batches, made by generate_signals.py)	Team-generated	Synthetic	Team-created, released with the project under Apache 2.0	N
Course list: courses.json (12 free courses, checked by hand on 2026-10-24)	Public course pages on Coursera, Khan Academy, HubSpot Academy, Kaggle Learn and edX	Self-collected (titles, links and hours only)	Course content belongs to each provider; only links and basic facts are stored	N
SDG mapping and skill-to-task map: sdg_mapping.json, skill_task_map.json (team drafts)	UN SDGs https://sdgs.un.org/goals and team analysis	Self-created	Team-created, released under Apache 2.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Test CVs (2 fictional CVs for testing and the demo)	Team-created	Synthetic	Team-created	N (all names and contact details are fake)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Anthropic Claude (claude.ai)	Interaction design, code writing, testing, and drafting documentation (README, demo script, test CVs, this declaration draft)	Moderate
Google Gemini (gemini.google.com)	Designing the LLM prompt for CV data extraction, defining JSON data schemas, and application logic structuring	Moderate
None	-	-
None	-	-
None	-	-

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

A team member read and ran every AI-generated change before committing it to GitHub. The app was tested end to end, locally and on Replit, through all five steps. The fairness test (14 cases) checks that personal details do not change rankings. Facts were checked by hand: the 12 courses were opened and verified on 2026-10-24, ESCO IDs were checked with scripts, and trend data is clearly labelled as synthetic. The team edited all documents and takes full responsibility for the submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud platform credits (USD 25)	Used to deploy and host the live prototype (Autoscale deployment) at https://sdc-hackathon-2026-team-160.replit.app, and to store the Gemini API key in Replit Secrets.
None	-	-
None	-	-
None	-	-

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not essential to the core logic. Replit hosts the live prototype, but the app is a standard Streamlit app with a requirements.txt and can run on any Python host with small changes. Gemini is used only to read CVs; it could be replaced by another LLM API by changing one function, and users without a CV can use the job templates even when the API is unavailable. RunPod was not used.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No accounts and no database. The app does not ask for name, age, gender or other demographic details. If a user uploads or pastes a CV, the text is processed in server memory: emails, phone numbers and Malaysian IC numbers are removed with regular expressions, and only the cleaned text is sent to the Google Gemini API. The profile lives only in the user's browser session and is lost when the session ends; the app does not write it to disk. Users save progress by downloading their own profile JSON and can re-upload it later. The API key is stored in Replit Secrets, not in the code or repository. Limitation: redaction is pattern-based, so names, addresses or unusual number formats in a CV may still reach Gemini. On the Gemini free tier, Google may use submitted content to improve its products.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini may misread a CV or guess the wrong skill level. Mitigation: Step 2 shows the evidence and confidence for every skill, and the user must review and edit before matching; skill names must reach a fuzzy-match score of 80 to count as ESCO skills. PII removal is pattern-based and can miss formats. The app depends on one AI provider and its rate limits; it retries once, and the job templates work without AI. Trend data is synthetic, and the skill-to-task map, SDG mapping and required skill levels are team drafts, so all results are labelled as estimates. Bias: the matching formula uses no personal attributes, and a 14-case fairness test confirms rankings do not change. Only 12 courses and an English interface for now. The google-generativeai SDK is deprecated and should move to google-genai.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Date: 25 October 2026